This is one of the possible solutions:

1. Null hypothesis:

   $$H_0: \mu_{TK35} \leq \mu_{PYV123}$$

   Alternative hypothesis:

   $$H_1: \mu_{TK35} > \mu_{PYV123}$$

   Where μ is the mean viral titers in the respective group. We would like to show that our experimental antiviral PYV123 reduces the viral titer in the lungs compared to TK35. Hence, we want to reject the hypothesis that the mean viral titer measured using TK35 is smaller than the mean viral titer measured using PYV123.

   Test to be used: one-sided two sample t-test (since we do not know the variance)

2. Significance level: anything reasonable, I take 0.05
3. Effect size = 1.5. This is the minimal clinically relevant effect.
4. Other parameters (variance): from pilot study SD ≈ 0.82. I will be somewhat conservative and take SD = 1.0
5. Power: I assume 80%, which is reasonable for discovery trials

Resulting total sample size is:

power.t.test(n=, delta=1.5, sig.level=0.05, sd=1.0, power=0.80, type =" two.sample", alternative=" one.sided", strict=TRUE)

7 per group, i.e. 14 in total

NOTE: This is a possible solution. Other choices will result in a different sample size.

Typical mistakes and comments:

A) **Inflation of variance.** Several of you inflated the variance observed in pilot study. It is good thing to do definitely! However, for few of you, reasoning behind seemed to be rather unclear, saying "for a larger experiment we expect larger variance". In general, we do not expect that a larger experiment will lead to a larger variance. The theoretical variance behind it is still the same, both in the pilot study and the main experiment, because both data are coming from the same distribution. The problem is in the precision of the estimate of this variance: the pilot study is usually small and therefore its precision can be rather bad and the estimate can be far from actual value of variance. Therefore a small inflation is good practice, because it protects us from underestimation of variance in sample size calculations. On the other hand, it should not be exaggerated neither, because pilot study can also overestimate

variability and further inflation makes it even worse in terms of unnecessary high resulting sample size.

B) **Z test versus t-test.** Strictly speaking, Z-test should not be used in this setting, because we do not know the variance ahead in our main experiment. We would thus use a t-test. Using the Z-test as main test is therefore completely wrong.
However, it can be used in sample size calculation, especially in case if you would not have a PC with you and you need to conduct experiment by hand. It would work to make calculations based on Z-test (that you can compute easily by hand) and then use t-test in your main experiment. If resulting N is high, approximation of t-test by Z-test could be pretty much ok, as we mentioned during the class. However, it is always slightly underestimate necessary sample size! (though in practice, the resulting number of subjects can be same, after rounding up)

C) **Few people mis-specified the directionality of the hypothesis!**

$$H_0: \mu_{TK35} \geq \mu_{PYV123}$$

This is essentially wrong and an important mistake, because all the results and interpretation will be wrong. Luckily enough, the results of the sample size calculation would not change, because it would come up exactly same. **So, remember: what you want to prove is always in the alternative!** Well, there are few very obscure exemptions, like tests for normality...

D) **Equality in the null hypothesis.**

$$H_0: \mu_{TK35} = \mu_{PYV123}$$

I would advise you to always write the null hypothesis in terms of inequality, when doing one-sided test. Although it will have no impact on your computations nor the analysis, it is much clearer for interpretation.

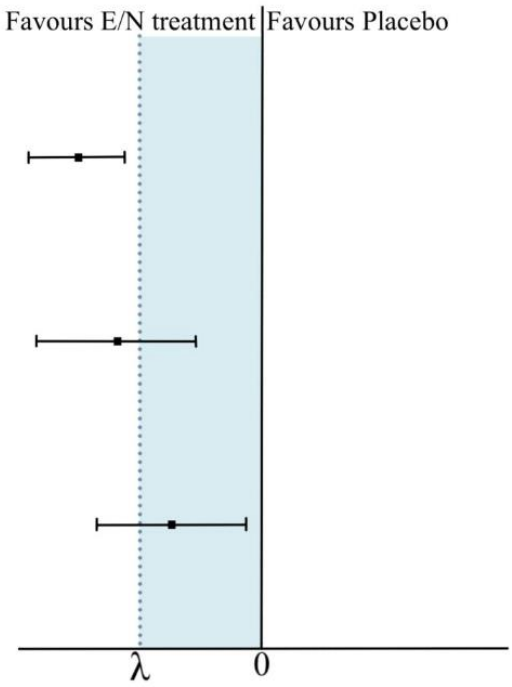
E) **Putting effect size into hypothesis.**
Null hypothesis:

$$H_0: \mu_{TK35} - \mu_{PYV123} \leq 1.5$$

Alternative hypothesis:

$$H_A: \mu_{TK35} - \mu_{PYV123} > 1.5$$

There were some students who wanted to test this hypothesis. It is not generally wrong to do this, there exist a whole theory about doing these types of tests and it is typical approach in some areas. However, you must adjust for this in sample size calculation. This is called superiority testing by a margin.
In the figure below, there is an example of a hypothesis with $H_0: \mu_2 - \mu_1 \geq \lambda \ vs \ H_A: \mu_2 - \mu_1 < \lambda$. All three 95% confidence intervals exclude zero, and thus when can conclude in all 3 cases that $\mu_2 < \mu_1$. However, only the top 95% confidence interval excluded the value of λ (the margin) to the left, and only with this confidence interval we are able to reject $H_0: \mu_2 - \mu_1 \geq \lambda$.



A) Superiority Trial

Figure from https://s4be.cochrane.org/blog/2022/03/18/understanding-non-inferiority-trials/